Relationship between variables

1 State whether there is likely to be a relationship between the following pairs of variables:

a Study time and test score.

b Eye colour and IQ.

c Typing speed and dancing ability.

d Driving time and petrol used.

2 For each pair of variables, state whether one variable has a direct effect on the other. If yes, state which variable affects the other.

a Arm length (cm), Weight (kg)

b Temperature $(^{\circ}\mathrm{C})$, Distance from the equator (km)

c Weekly income, Number of friends

d Time of travel (minutes), Distance covered (m)

e Age (years), Time spent watching television

f Time spent working, Wages earned

g Time spent to finish a novel, Number of pages

h Marital status, Film preference

i Height (cm), Number of children

j Hair Length, Gender

k Number of pets, Time spent caring for pets

l Time spent training, Athletic performance

3 For each of the following pairs of variables:

i State the independent variable.

ii State the dependent variable.

a Number of ice cream cones sold and temperature outside

b Size of a pizza and cost of a pizza

c Time spent studying and exam mark

4 Consider the following variables:

- Ticket sales
- Revenue from a show

a Which variable affects the other?

b State the dependent and independent variables.

5 A kite can rise to a height of $161 \mathrm{~m}$ when the wind is blowing at 6 kilometres per hour and to a height of $322 \mathrm{~m}$ when the wind is blowing at 12 kilometres per hour.

a State the two variables in the above scenario.

b From the two variables, state the dependent and independent variables.

6 A study was performed to find the relationship between the number of soft drinks consumed per week and the waist circumference of students.

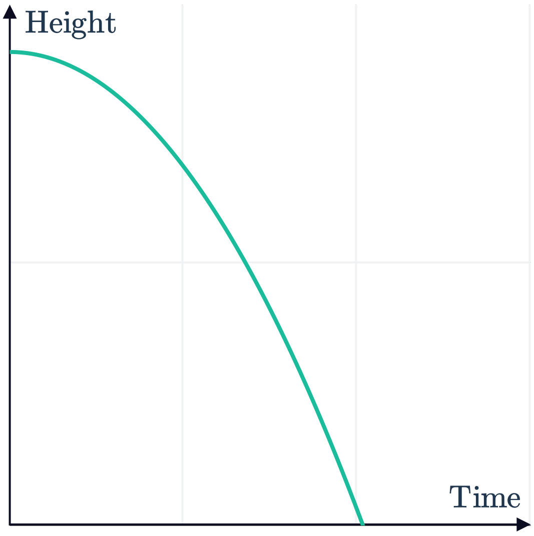
a Which variable is the independent variable?

b Which variable is the dependent variable?

c Which variable should be put on the x-axis?

d Which variable should be put on the y-axis?

7 The graph shows the height of a ball after it is dropped off the side of a building:



a Which variable is the dependent variable?

b Which variable is the independent variable?

8 Describe the relationship between the variable[redacted]he following tables:

a Eight students' marks in English and History

Mark in English	77	81	65	61	97	57	88	71
Mark in History	51	65	71	80	72	62	86	92

b GDP per capita and infant mortality rate of eight countries

GDP per capita (USD)	2100	3700	3900	3500	9200	3300	8400	3900
Infant mortality rate	80	36	15	42	13	46	2	35

cHeight and weight of ten people

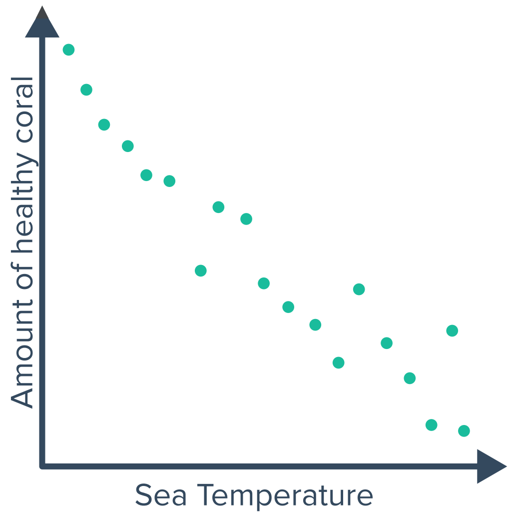
Height (cm)	194	187	166	184	165	162	174	190	163	163
Weight (kg)	75.27	73.43	52.36	60.94	62.62	52.49	57.52	72.20	53.14	61.11

Scatter plots

9 The scatter plot shows the relationship between sea temperature and the amount of healthy coral:

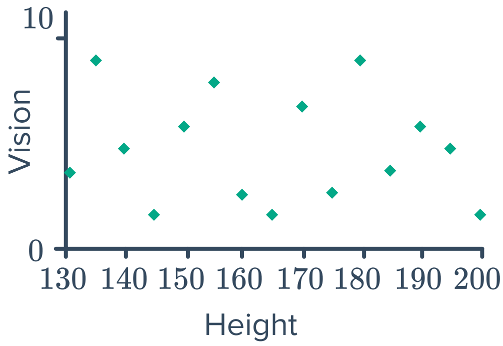
a Which variable is the dependent variable?

b Which variable is the independent variable?

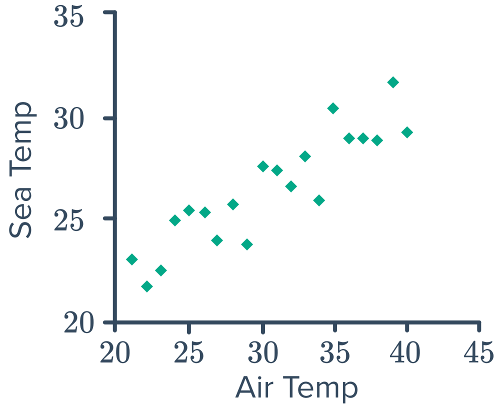


10 Describe the relationship between the variables in the following scatter plots:

a

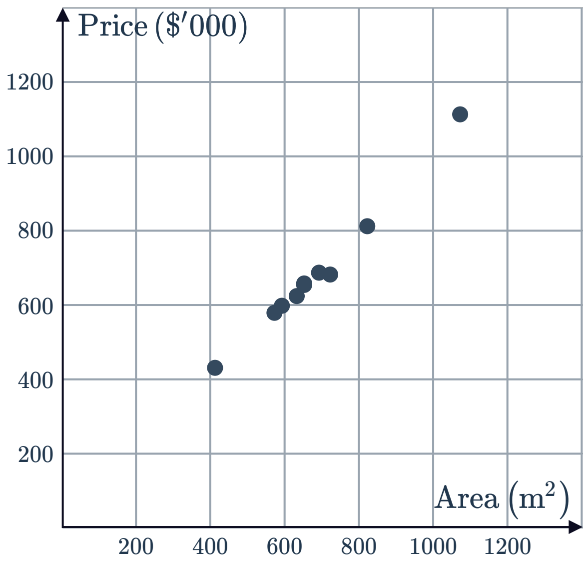


b

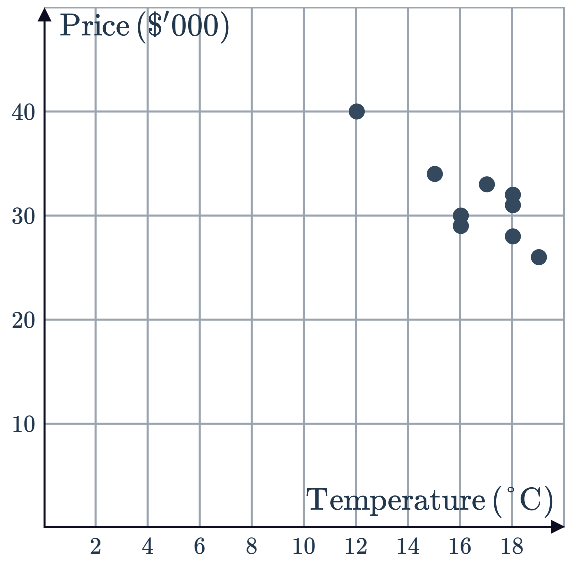


11 Describe the relationship between the variables in the following scenarios:

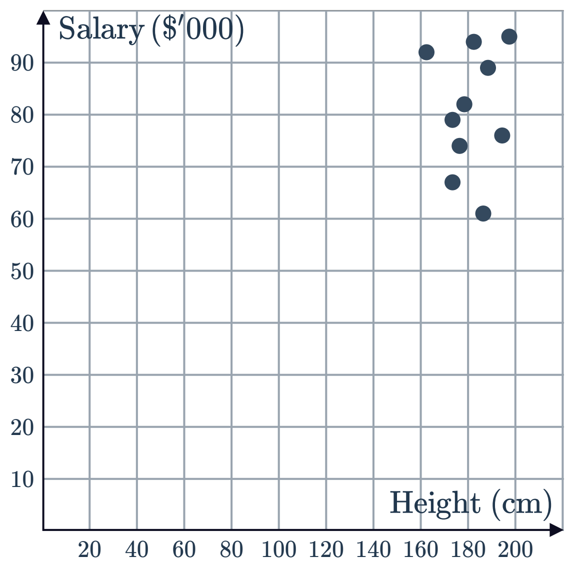
a The price of ten houses is graphed against the house's land area.

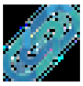


b A cafe records the number of soups sold and the daily maximum temperature.



c A company records the height and annual salary of ten employees.



12 Construct a scatter plot for the set of data  the following table:

x	1	3	5	7	9
y	3	7	11	15	19

13 The table shows the number of apples eaten per week and the number of visits to the doctor per year by six students:

Number of apples eaten per week	0	3	6	7	9	11
Number of visits to the doctor per year	17	14	11	6	3	0

a Construct a scatter plot for this data.

b Vanessa wants to visit the doctor as few times as possible each year. How many apples should she eat per week?

c Is this prediction reliable? Explain your answer.

14 The table shows the age and height of six children in a family:

Age (years)	3	4	9	11	13	16
Height (cm)	90	105	130	145	160	165

a Construct a scatter plot for this data.

b Bob is the second youngest child in the family. How tall is Bob?

15 The table shows the number of books read per year by six students and the mark they received in their yearly maths exam:

Number of books read per year	5	15	20	30	40	50
Mark in yearly maths exam	50	65	70	80	90	100

a Construct a scatter plot for this data.

b In this group of students, Valentina achieved the highest mark in the yearly maths exam. How many books did she read throughout the year?

c The teacher told the class: "The more books you read, the higher your yearly maths exam mark will be." Based on this data, is the teacher correct?

16 The table shows the number of carrots eaten per day by six rabbits and their hop length:

Number of carrots eaten per day	2	3	6	10	14	16
Hop length (cm)	43	70	46	54	51	59

a Construct a scatter plot for this data.

b Find the difference in hop length between the rabbit who eats the most carrots and the rabbit who eats the least carrots per day.

17 The number of customers that visit a new clothing store is recorded on six different days as shown in the table:

Number of days since store opened	1	2	3	5	7	9
Number of customers	10	15	25	35	35	60

a Construct a scatter plot for this data.

b Find the average number of customers that went to the store for the six days.

18 Construct a scatter plot for the sets of data [illegible]e following scenarios:

a An experiment was conducted and the following table shows the results:

x	1	3	6	8	9	11	14	15	17	20
y	2	4	6	8	12	24	30	46	52	64

b A computer program generates ten random coordinate pairs to animate a display:

x	16	14	7	15	13	8	10	7	20	5
y	18	16	16	5	6	1	18	12	1	10

c A student was performing an experiment to study the relationship between the current and voltage through a resistor. He noted his results in the following table:

Current (x)	1	2	3	4	5	6	7	8	9	10
Voltage (y)	5	14	15	26	33	34	44	47	57	57

d The following table shows the time taken to finish a lap on a race track for several average speeds:

Speed (x)	20	25	30	35	40	45	50	55	60	65
Time (y)	89	89	79	77	72	68	57	64	46	45

e A study was conducted to find the relationship between the age at which a child first speaks and their level of intelligence as teenagers. The following table shows the ages of some teenagers (in months) when they first spoke, and their results in an aptitude test:

Age when first spoke (x)	14	25	11	9	13	20	18	10	7
Aptitude test results (y)	95	70	85	89	101	86	93	100	103

f Ten runners in a park record the distance (in kilometres) they run and the amount of time (in minutes) they spend running:

Time (x)	28	11	37	26	43	37	41	47	19	43
Distance (y)	9.3	3.6	10.8	6.6	13.8	12	11.7	15	6.3	13.2

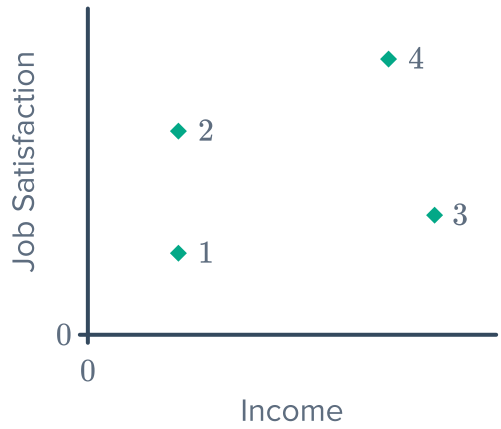
19 The height of the tide at a beach is record[redacted]ten different hours in one day:

Time (hours)	13	24	7	2	4	15	6	11	21	10
Height (m)	-0.21	-0.05	0.92	0.55	0.84	-0.76	1.01	0.31	-0.68	0.5

a State the dependent and independent variables.

b Construct a scatter plot for this data.

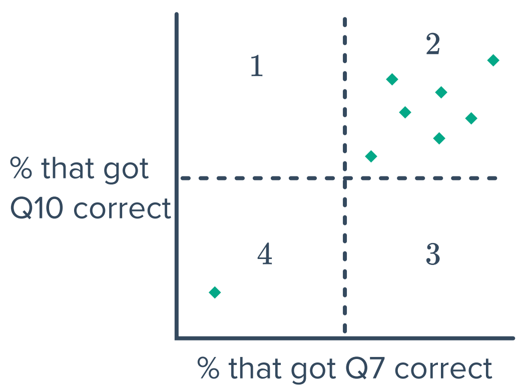
20 Georgia has a high income but does not like her job.
Which dot is most likely to represent Georgia?



21 After a mathematics exam, Lachlan commented that he left out Question 10 because it was not relevant to the topics that were assessed, but that the other questions were fine. The teacher felt that Question 10 and Question 7 were assessing the same skills, so she decided to look at the results of students across all 8 classes. She plotted the percentage who had gotten Question 10 correct against the percentage who had gotten Question 7 correct as shown below:

a According to the graph, who is correct: Lachlan or the teacher?

b Let's assume that Lachlan performed exceptionally well in every other question. If a whole class of students had the same result as Lachlan, where could this class's results appear on the scatter plot, region 1, 2, 3 or 4?



22 Consider the following table showing the number of times per day that college students check Facebook and the amount of time they spend in preparing for class:

Number of Facebook Checks	Time preparing for class (minutes)
1	100
2	90
5	80
8	40
11	50
12	60
13	30
16	20
19	10

a State the independent variable.

b Which axis should the independent variable appear on?

c State the dependent variable.

d Which axis should the dependent variable appear on?

e Construct a scatter plot for the data.

f Describe the type of relationship shown in your scatter plot.

23 The number of accidents on a highway and the amount of rainfall on ten rainy days are recorded in the table below:

Amount of rainfall (mm)	1	17	17	18	17	24	4	6	23	1
Number of accidents	8	15	16	15	14	19	11	9	17	9

a State the independent and dependent variables.

b Construct a scatter plot using the data from the table.

c Describe the relationship between rainfall and the number of accidents.

24 The volume at different pressures of a gas are measured and recorded in the table below:

Pressure (kPa)	400	500	1600	1200	100	800	1100	1300	1500	1800
Volume (cm^3)	2.2	1.6	0.4	0.7	8	0.9	0.9	0.5	0.7	0.3

a State the independent and dependent variables.

b Construct a scatter plot using the data from the table.

c Describe the relationship between pressure and volume.

25 Scientists were looking for a relationship between the number of hours of sleep we receive and the effect it has on our motor and process skills. Some subjects were asked to sleep for different amounts of time, and were all asked to undergo the same driving challenge in which their reaction time was measured. The data is recorded in the table below:

Amount of sleep (hours)	9	6	4	10	3	7	2	5
Reaction time (seconds)	3	3.3	3.5	3	3.7	3.2	3.85	3.55

a Construct a scatter plot using the data from the table.

b Is amount of sleep and reaction time negatively correlated, postively correlated or not correlated?

c According to the results, does an increased amount of sleep generally lead to improved reaction times?

26 To determine whether the presence of sharks in a coastal region is influenced by cage diving, the number of nearby cage diving operations and the number of nearby shark sightings was recorded each month over several months. The results are shown in the table below:

Cage Diving Operations	1	9	7	6	3	4	2	5	8	10
Shark Sightings	5	8	4	9	10	3	3	3	1	5

a One observation in the data set is $(8, 1)$. Explain what this point represents in context.

b Construct a scatter plot using the data from the table.

c Describe the relationship between the number of shark sightings and the number of cage diving operations.

d According to the data, is it possible to determine whether cage diving operations encourage more sharks to come near the shoreline?